

Computational Fluency Short Course

Architectures and interfaces

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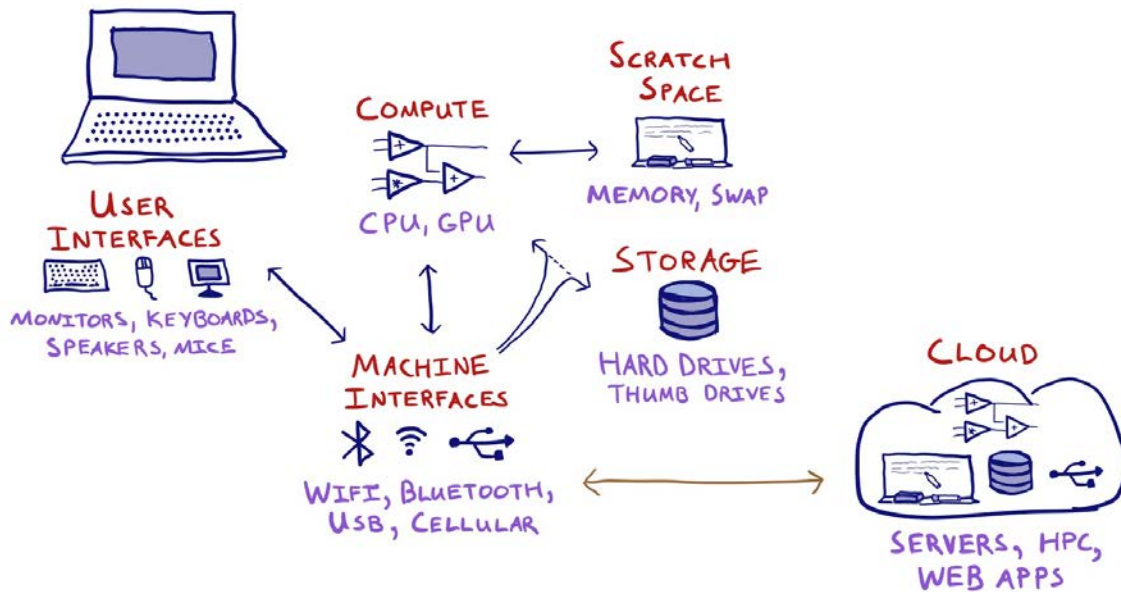
Scientific Director of Quantitative Neuroscience



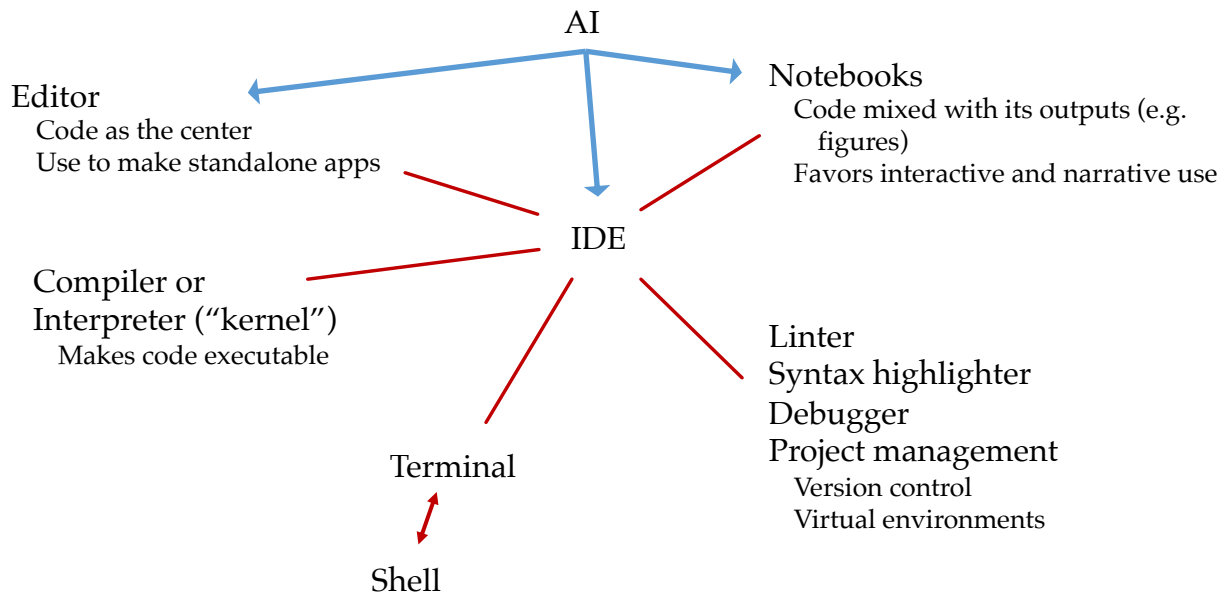
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<https://github.com/brownridd/cfsc26>

Back to basics: What is a computer?



Ways to interact with a computer



Which is the right interface for the job?



There are **many** other tools for computational projects, and everyone has their own preferred tool chain.

Common use cases:

- CLI - Direct interaction with the OS, processes, and filesystem
- Text editor - "Simple" files like scripts, READMEs, and configuration files
- Notebook - Exploratory data analysis, and "narrative" coding
- IDE - Exploratory data analysis, and "standalone" or complex coding

Do not use Excel:

nature

NEWS | 13 August 2021 | Correction 25 August 2021

Autocorrect errors in Excel still creating genomics headache

Despite geneticists being warned about spreadsheet problems, 30% of published papers contain mangled gene names in supplementary data.

<https://www.nature.com/articles/d41586-021-02211-4>